

The Process Architecture Principle

A Structural Extension of Green-Function Architecture Theory Toward Particle Reactions and Decays

Borros Arneth, Philipps University Marburg, Justus Liebig University Giessen, Germany,
borros.arneth@staff.uni-marburg.de

Abstract

Green functions provide one of the central mathematical languages of theoretical physics, describing propagation, response, and transport in classical field theory, quantum mechanics, many-body theory, and quantum field theory. Feynman diagrams provide a complementary graphical language for perturbative interactions, in which propagators and vertices generate transition amplitudes for particle reactions and decays. However, neither formalism introduces an explicit primitive notion of a complete process architecture. This manuscript proposes the **Process Architecture Principle**, according to which every physical reaction or decay possesses an associated **Process Green-Function Architecture**. This architecture is decomposed into four primitive structural sectors: propagation, interaction, holonomy, and recursion. The first two correspond closely to the conventional propagators and vertices of Feynman diagrammatics, while the latter two make explicit structural features that are usually encoded only implicitly through loops, topology, self-energy insertions, and recursive diagrammatic expansions. The manuscript introduces the **Architectural Multiplicity of a Process**, Π_{arch} , and develops structural analogues of transition amplitudes and decay widths. Preliminary comparisons with particle decays suggest that particle multiplicity is relevant for rest energies and decay Q -values, whereas process multiplicity is the natural object associated with transition strengths. The product/educt multiplicity quotient is discussed and rejected as a universal amplitude factor, but reinterpreted as a possible architectural restructuring index.

Keywords: Green functions; Feynman diagrams; particle decays; transition amplitudes; Green-Function Architecture Theory; process architecture; holonomy; recursion.

1. Introduction

Green functions are fundamental objects in theoretical physics. Since Green's original work on potential theory [1], they have become central in boundary-value problems, partial differential equations, spectral theory, quantum mechanics, many-body theory, and quantum field theory [2–10]. In their most general form, Green functions describe how a localized source produces a response elsewhere in a system.

Feynman diagrams introduced another essential layer. They encode perturbative processes through propagator lines, interaction vertices, loops, and diagrammatic expansions [11–15]. They are not merely pictures; they correspond to algebraic terms in perturbation theory and provide a compact method for computing scattering amplitudes, decay amplitudes, loop corrections, self-energies, and renormalized quantities.

Green functions and Feynman diagrams are therefore deeply connected. Green functions describe propagation, while Feynman diagrams organize propagation and interaction into process amplitudes. Yet neither formalism explicitly treats the complete physical process as a primitive architectural object.

Green-Function Architecture Theory (GFAT) proposes such a structural layer. Earlier GFAT work introduced the idea that particles may possess intrinsic Green-function architectures whose structural multiplicity is related to their rest energy [16-25]. The present manuscript extends this idea from static particle architectures to dynamic process architectures.

The central question is:

Can a particle reaction or decay be described as a Green-function architecture in its own right?

The answer proposed here is yes. Every physical process is associated with a **Process Green-Function Architecture**

$$\mathcal{A}_{\text{process}} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}),$$

where $\mathcal{P}$ denotes propagation, $\mathcal{I}$ interaction, $\mathcal{H}$ holonomy, and $\mathcal{R}$ recursion.

The manuscript develops this principle and shows how it creates a direct structural analogy with Feynman diagrammatics.

2. Green Functions as Transport Architectures

2.1 Green functions

A Green function is usually defined as the kernel of an inverse operator. If L is a differential or quantum operator, then the Green function G satisfies

$$LG = \delta.$$

Thus $G(x, y)$ describes the response at x generated by a source at y . This interpretation appears in potential theory, wave propagation, heat conduction, quantum mechanics, many-body theory, and transport theory [2–10].

Green functions therefore naturally encode propagation. They answer the question:

If something is inserted here, what response appears there?

In quantum field theory, propagators are Green functions. They describe the propagation of virtual particles between interaction vertices [11–15].

2.2 Green-function architectures

GFAT adds a structural layer beneath the analytical Green function. The Green function is not regarded as the primitive object, but as the analytical representation of an underlying transport architecture.

A Green-function architecture may be schematically written as

$$\mathcal{A} = (\Gamma, \mathcal{L}, \mathcal{B}, \mathcal{I}, \mathcal{R}),$$

where Γ is a transport graph, $\mathcal{L}$ a family of local operators, $\mathcal{B}$ boundary or domain data, $\mathcal{I}$ interface or interaction structure, and $\mathcal{R}$ recursive closure.

In this view,

$$\mathcal{A} \rightarrow G_{\mathcal{A}}.$$

The architecture determines the Green-function representation.

3. The Feynman Diagram Method

Feynman diagrams provide a graphical representation of terms in perturbative quantum field theory [11–15]. Each diagram corresponds to a mathematical expression contributing to a transition amplitude.

The basic graphical elements are:

- external lines,
- internal propagator lines,
- interaction vertices,
- loops,

- repeated insertions.

A propagator line corresponds to a Green function. A vertex corresponds to an interaction term in the Lagrangian. A complete diagram corresponds to an amplitude contribution.

For example, a process

$$i \rightarrow f$$

is described by a transition amplitude

$$\mathcal{M}_{i \rightarrow f}.$$

Observable quantities such as cross sections and decay widths are then obtained from

$$|\mathcal{M}|^2$$

combined with phase-space integration.

The strength of Feynman diagrammatics is that it provides an operational method for calculating physical processes. However, its elementary graphical primitives are primarily propagation lines and interaction vertices. Holonomy and recursion are present, but usually appear implicitly through loops, self-energy insertions, Dyson equations, and renormalization [12–18].

This observation motivates the GFAT extension.

4. The Process Architecture Principle

The fundamental principle proposed here is the following.

Process Architecture Principle. Every physical reaction or decay possesses an associated Process Green-Function Architecture

$$\mathcal{A}_{\text{process}} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}).$$

The four primitive sectors are:

$$\begin{aligned} \mathcal{P} &= \text{propagation,} \\ \mathcal{I} &= \text{interaction,} \end{aligned}$$

$\mathcal{H}$ = holonomy,
 $\mathcal{R}$ = recursion.

Propagation describes transport along admissible channels. Interaction describes coupling at vertices or interfaces. Holonomy describes closed transport, loops, and global phase structure. Recursion describes repeated insertion, self-energy correction, Dyson-type closure, and self-consistent process generation.

Thus the process is not described only by what enters and what exits. It is described by the architecture through which the transition occurs.

5. Static versus Dynamic Green-Function Architectures

GFAT distinguishes two architectural levels.

First, a **particle architecture** describes the intrinsic structure of a stable or quasi-stable particle. Its Architectural Multiplicity is denoted

$$\Pi_{\text{particle}}.$$

In the static version of GFAT, particle rest energy is proposed to satisfy

$$E_X = E_{\text{Ryd}} \Pi_X + E_{\text{corr}},$$

where E_{corr} may include charge, isospin, electromagnetic, weak, or other correction terms [25].

Second, a **process architecture** describes a reaction or decay. Its multiplicity is denoted

$$\Pi_{\text{arch}}.$$

The distinction is essential:

$$\Pi_{\text{particle}} \rightarrow \text{rest energy},$$

whereas

$$\Pi_{\text{arch}} \rightarrow \text{transition strength}.$$

Thus, a particle architecture determines intrinsic energy, while a process architecture determines dynamic transition behavior.

6. The Analogy with the Feynman Diagram Method

This section gives the central comparison.

The Feynman diagram method and the Process Architecture Principle describe the same physical domain, but at different structural levels.

Feynman Diagram Method	GFAT Process Formalism
External particles	Boundary conditions of the process
Propagator	Propagation architecture $\mathcal{P}$
Interaction vertex	Interaction architecture $\mathcal{I}$
Closed loop	Holonomy architecture $\mathcal{H}$
Self-energy-insertion, Dyson resummation	Recursion architecture $\mathcal{R}$
Complete Feynman graph	Complete Process Green-Function Architecture
Transition amplitude $\mathcal{M}$	Architectural transition amplitude $\mathcal{M}_{\text{GFAT}}$

The first two GFAT sectors are already explicit in Feynman diagrams:

$$\begin{aligned} \text{propagator line} &\leftrightarrow \mathcal{P}, \\ \text{interaction vertex} &\leftrightarrow \mathcal{I}. \end{aligned}$$

The latter two are present but less explicit:

$$\begin{aligned} \text{closed loop} &\leftrightarrow \mathcal{H}, \\ \text{recursive expansion} &\leftrightarrow \mathcal{R}. \end{aligned}$$

This suggests the following correspondence principle.

Feynman–GFAT Correspondence Principle. Every perturbative Feynman graph determines a corresponding Process Green-Function Architecture. GFAT decomposes this architecture into propagation, interaction, holonomy, and recursion.

This statement does not imply that GFAT replaces Feynman diagrammatics. Rather, GFAT provides a structural interpretation and extension of the diagrammatic method.

7. Completing the Structural Decomposition

Feynman diagrams explicitly represent two primitive structures:

1. propagation,
2. interaction.

GFAT proposes that a complete architectural description of a process requires two additional primitive structures:

3. holonomy,
4. recursion.

Holonomy includes graph cycles, closed transport, Wilson loops, Berry-type phase structures, and loop topology [16–19]. Recursion includes self-energy insertions, Dyson equations, skeleton expansions, renormalization, and self-consistent closure [12],[20],[21].

The proposed four-sector decomposition is therefore

$$\mathcal{A}_{\text{process}} = \mathcal{P} \oplus \mathcal{I} \oplus \mathcal{H} \oplus \mathcal{R}.$$

This decomposition does not alter conventional amplitudes. Instead, it adds an explicit architectural classification to the process.

8. Architectural Multiplicity of a Process

For a process architecture, define the Architectural Multiplicity

$$\Pi_{\text{arch}} = 2^{a_p} 3^{b_p} (3\pi)^{c_p} A^{d_p/8}$$

where

$$A = \left(1 - \frac{1}{3\pi}\right)^{-1}$$

Here

$$a_p$$

measures the propagation sector,

$$b_p$$

the interaction sector,

$$c_p$$

the holonomy sector,

and

$$d_p$$

the recursion sector.

This is the process analogue of the particle Architectural Multiplicity introduced previously [25]. The important conceptual distinction is:

$$\Pi_{\text{particle}} \neq \Pi_{\text{arch}}.$$

The former belongs to a particle. The latter belongs to a reaction.

9. Process Transition Amplitudes

In standard quantum field theory, a Feynman diagram contributes to a transition amplitude

$$\mathcal{M}.$$

GFAT proposes an architectural analogue:

$$\mathcal{M}_{\text{GFAT}} = g_{\text{arch}} \Pi_{\text{arch}}.$$

Here g_{arch} is an interaction-dependent coupling scale, and Π_{arch} measures the complexity of the process architecture.

A previously considered possibility was to include the quotient

$$\frac{\Pi_{\text{products}}}{\Pi_{\text{educts}}}.$$

However, preliminary comparisons show that this quotient is not supported as a universal amplitude factor. In purely leptonic weak decays and in leptonic W decays, such a quotient would produce large artificial differences between channels that are experimentally known to be nearly universal [24].

Thus the quotient should not be included in the universal transition amplitude.

A more plausible interpretation is that

$$\frac{\Pi_{\text{products}}}{\Pi_{\text{educts}}}$$

may represent an **architectural restructuring index**, relevant in selected composite-particle processes where internal architectures are reorganized. It is not a universal dynamical factor.

Therefore the preferred first-order GFAT amplitude is

$$\mathcal{M}_{\text{GFAT}} = g_{\text{arch}} \Pi_{\text{arch}}$$

10. GFAT Decay Widths

In conventional quantum field theory, the decay width is schematically

$$\Gamma \sim |\mathcal{M}|^2 \rho(Q),$$

where $\rho(Q)$ is the phase-space factor.

The GFAT analogue is

$$\Gamma_{\text{GFAT}} = \Gamma_0 g_{\text{arch}}^2 \Pi_{\text{arch}}^2 \rho(Q) S_{\text{selection}}.$$

Here

$$\Gamma_0$$

sets the unit scale,

$$\mathcal{G}_{\text{arch}}$$

is the interaction-dependent coupling,

$$\Pi_{\text{arch}}$$

is the process Architectural Multiplicity,

$$\rho(Q)$$

is the phase-space factor,

and

$$\mathcal{S}_{\text{selection}}$$

collects selection effects such as spin, helicity, flavor structure, CKM factors, decay constants, hadronic overlap, and mixing.

For three-body leptonic weak decays, one expects approximately

$$\rho_3(Q) \sim Q^5.$$

For two-body decays, the phase-space factor is related to the final momentum. For pseudoscalar two-photon decays, the leading phase-space dependence is commonly proportional to a cubic power of the mass scale.

The essential GFAT separation is:

$$\begin{aligned} \Pi_{\text{particle}} &\rightarrow Q, \\ \Pi_{\text{arch}} &\rightarrow \text{transition strength}, \\ \rho(Q) &\rightarrow \text{kinematic availability}. \end{aligned}$$

11. Experimental Validation

The following comparisons are preliminary. They are intended to test whether the separation between particle multiplicity and process multiplicity is useful.

11.1 Rest energies

The earlier GFAT mass relation proposes

$$E_X = E_{\text{Ryd}} \Pi_X.$$

The uploaded GFAT particle-energy paper lists tuples and approximate energies for leptons, quarks, vector bosons, and hadrons, including the electron, muon, tau, W, Z, pion, kaon, eta, rho, omega, phi, proton, and sigma baryon [25].

These rest energies provide the basis for decay Q -value calculations.

11.2 Decay energies

For a decay

$$A \rightarrow B + C + \dots ,$$

GFAT predicts

$$Q_{\text{GFAT}} = E_A^{\text{GFAT}} - \sum_i E_i^{\text{GFAT}} .$$

Examples include:

$$\begin{aligned} \mu^- &\rightarrow e^- + \bar{\nu}_e + \nu_\mu, \\ \pi^0 &\rightarrow \gamma + \gamma, \\ K^+ &\rightarrow \mu^+ + \nu_\mu, \\ \tau^- &\rightarrow e^- + \bar{\nu}_e + \nu_\tau. \end{aligned}$$

Preliminary comparisons show that muon decay and neutral pion decay are close when the paper's GFAT rest energies are used. Tau and electroweak-scale cases show deviations requiring correction layers.

11.3 Leptonic lifetimes

For pure leptonic weak decays,

$$\begin{aligned}\mu &\rightarrow e\nu\nu, \\ \tau &\rightarrow e\nu\nu, \\ \tau &\rightarrow \mu\nu\nu,\end{aligned}$$

the same weak process architecture is expected. Thus

$$\Pi_{\text{arch}} = \Pi_W$$

is common, and the main difference comes from phase space.

After calibrating the weak constant on muon decay, the architecture-only law

$$\Gamma_{\text{GFAT}} \propto \Pi_W^2 \rho_3(Q)$$

gives promising relative results for tau leptonic decays. This supports the view that the product/educt quotient should not be included for elementary leptonic weak processes.

11.4 Hadronic decays

Hadronic decays are more complex because internal composite architectures may be reorganized.

Relevant examples include

$$\begin{aligned}\pi^+ &\rightarrow \mu^+ \nu_\mu, \\ K^+ &\rightarrow \mu^+ \nu_\mu, \\ \rho &\rightarrow \pi\pi, \\ K^* &\rightarrow K\pi, \\ \Delta &\rightarrow N\pi.\end{aligned}$$

Preliminary tests suggest that the product/educt quotient may help in selected hadronic-source comparisons, such as $K^+ \rightarrow \mu\nu$ relative to $\pi^+ \rightarrow \mu\nu$, but fails as a universal hadronic decay factor. Therefore, it should be treated, if at all, as part of $S_{\text{selection}}$, not as part of the universal amplitude.

The current preferred formula remains

$$\Gamma_{\text{GFAT}} = \Gamma_0 g_{\text{arch}}^2 \Pi_{\text{arch}}^2 \rho(Q) S_{\text{selection}}.$$

12. Process Architectures Beyond Feynman Diagrams

The Process Architecture Principle is not limited to perturbative Feynman diagrams. It suggests a broader structural language for physical processes.

Possible extensions include:

- non-perturbative Green functions,
- lattice gauge theory,
- Dyson–Schwinger systems,
- topological phases,
- Wilson-loop structures,
- quantum gravity.

In perturbative quantum field theory, Feynman diagrams provide a natural representation of process architectures. In non-perturbative settings, the process architecture may not be representable by a finite diagram. Nevertheless, the four-sector decomposition may remain meaningful:

$$\mathcal{P} \oplus \mathcal{I} \oplus \mathcal{H} \oplus \mathcal{R}.$$

This suggests that GFAT may provide a bridge between perturbative diagrammatics and non-perturbative structural approaches.

13. Discussion

The present manuscript proposes a dynamic extension of GFAT.

The main conceptual advance is the distinction between particle architectures and process architectures. Particle architectures are associated with intrinsic rest energies. Process architectures are associated with transition strengths.

The analogy with Feynman diagrams is central. Feynman diagrams already encode propagation and interaction explicitly. GFAT proposes that holonomy and recursion should also be treated as primitive architectural sectors.

This does not invalidate Feynman diagrammatics. On the contrary, it places GFAT in direct continuity with it. Feynman diagrams remain the standard perturbative computational tool. GFAT provides an additional structural interpretation of the same process.

The strengths of the present framework are:

1. It gives a structural interpretation of reactions.

2. It distinguishes static and dynamic architectures.
3. It connects directly to Feynman diagrammatics.
4. It produces testable decay-energy and lifetime relations.
5. It identifies the product/educt quotient as non-universal.

The limitations are also clear:

1. The process multiplicity Π_{arch} must be rigorously derived.
2. Selection factors require a systematic theory.
3. Hadronic decays need a more complete treatment of composite architecture.
4. The mapping from arbitrary Feynman diagrams to GFAT architectures remains to be formalized.
5. Experimental validation remains preliminary.

14. Conclusions

This manuscript introduced the **Process Architecture Principle**, according to which every particle reaction or decay possesses an underlying Process Green-Function Architecture.

The main conclusions are:

1. GFAT can be extended from static particle architectures to dynamic process architectures.
2. The framework provides a structural extension of Green-function theory.
3. The analogy with the Feynman diagram method is direct and central.
4. Propagation and interaction correspond to the standard propagators and vertices of Feynman diagrams.
5. Holonomy and recursion are proposed as additional primitive architectural sectors.
6. The Architectural Multiplicity of a Process, Π_{arch} , provides a natural structural analogue of transition strength.
7. The leading GFAT decay-width formula is

$$\Gamma_{\text{GFAT}} = \Gamma_0 g_{\text{arch}}^2 \Pi_{\text{arch}}^2 \rho(Q) S_{\text{selection}}.$$

8. The product/educt quotient is not supported as a universal amplitude factor.
9. It may instead represent an architectural restructuring index in selected composite processes.
10. Future work should develop a rigorous Feynman–GFAT mapping and perform large-scale validation against decay-width data.

Declarations

Funding

No external funding was received for this work.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability

No new experimental datasets were generated. Particle data used for comparison are available from the Particle Data Group [24]. GFAT particle-energy values and exponent tuples are taken from the author's prior manuscript [25].

Author Contributions

The author conceived the theoretical framework, developed the process architecture principle, performed the conceptual analysis, and wrote the manuscript.

References

1. Green G. An Essay on the Application of Mathematical Analysis to the Theories of Electricity and Magnetism. Nottingham: T Wheelhouse; 1828.
2. Courant R, Hilbert D. Methods of Mathematical Physics. Vol I. New York: Wiley-Interscience; 1953.
3. Stakgold I, Holst M. Green's Functions and Boundary Value Problems. 3rd ed. Hoboken: Wiley; 2011.
4. Evans LC. Partial Differential Equations. 2nd ed. Providence: American Mathematical Society; 2010.
5. Reed M, Simon B. Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness. New York: Academic Press; 1975.
6. Economou EN. Green's Functions in Quantum Physics. 3rd ed. Berlin: Springer; 2006.
7. Fetter AL, Walecka JD. Quantum Theory of Many-Particle Systems. New York: McGraw-Hill; 1971.
8. Mahan GD. Many-Particle Physics. 3rd ed. New York: Kluwer Academic/Plenum; 2000.
9. Abrikosov AA, Gorkov LP, Dzyaloshinskii IE. Methods of Quantum Field Theory in Statistical Physics. New York: Dover; 1975.
10. Kadanoff LP, Baym G. Quantum Statistical Mechanics. New York: Benjamin; 1962.
11. Feynman RP. Space-time approach to quantum electrodynamics. Phys Rev. 1949;76:769–789.

12. Dyson FJ. The S matrix in quantum electrodynamics. *Phys Rev.* 1949;75:1736–1755.
13. Peskin ME, Schroeder DV. *An Introduction to Quantum Field Theory*. Reading: Addison-Wesley; 1995.
14. Weinberg S. *The Quantum Theory of Fields. Vol I*. Cambridge: Cambridge University Press; 1995.
15. Itzykson C, Zuber JB. *Quantum Field Theory*. New York: McGraw-Hill; 1980.
16. Wilson KG. Confinement of quarks. *Phys Rev D.* 1974;10:2445–2459.
17. 't Hooft G. A planar diagram theory for strong interactions. *Nucl Phys B.* 1974;72:461–473.
18. Mandelstam S. Feynman rules for electromagnetic and Yang-Mills fields from the gauge-independent field-theoretic formalism. *Phys Rev.* 1968;175:1580–1603.
19. Berry MV. Quantal phase factors accompanying adiabatic changes. *Proc R Soc Lond A.* 1984;392:45–57.
20. Luttinger JM, Ward JC. Ground-state energy of a many-fermion system. II. *Phys Rev.* 1960;118:1417–1427.
21. Baym G. Self-consistent approximations in many-body systems. *Phys Rev.* 1962;127:1391–1401.
22. Datta S. *Electronic Transport in Mesoscopic Systems*. Cambridge: Cambridge University Press; 1995.
23. Freericks JK. *Transport in Multilayered Nanostructures: The Dynamical Mean-Field Theory Approach*. London: Imperial College Press; 2006.
24. Workman RL, Burkert VD, Crede V, et al. Review of Particle Physics. *Prog Theor Exp Phys.* 2022;2022:083C01.
25. Arneth B. A New Method for Calculating the Energies Associated with Particle Reactions. Preprint manuscript; 2025.

Appendix A. Mathematical Foundations

Let a process be denoted

$$i \rightarrow f.$$

A process architecture is defined as

$$\mathcal{A}_{\text{process}} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}).$$

The four projection maps are

$$\begin{aligned} P(\mathcal{A}) &= \mathcal{P}, \\ I(\mathcal{A}) &= \mathcal{I}, \\ H(\mathcal{A}) &= \mathcal{H}, \end{aligned}$$

$$R(\mathcal{A}) = \mathcal{R}.$$

The process coordinate tuple is

$$(a_p, b_p, c_p, d_p).$$

The corresponding multiplicity is

$$\Pi_{\text{arch}} = 2^{a_p} 3^{b_p} (3\pi)^{c_p} A^{d_p/8}.$$

The decay width ansatz is

$$\Gamma_{\text{GFAT}} = \Gamma_0 g_{\text{arch}}^2 \Pi_{\text{arch}}^2 \rho(Q) S_{\text{selection}}.$$

Appendix B. Conditional Proofs

Theorem B1. Existence of a reduced process architecture

If a process admits a graph representation with propagators, vertices, cycles, and recursive insertions, then it admits a reduced GFAT process architecture.

Proof.

Propagators define $\mathcal{P}$. Vertices define $\mathcal{I}$. Cycles define $\mathcal{H}$. Recursive insertions define $\mathcal{R}$. Therefore the process determines

$$\mathcal{A}_{\text{process}} = (\mathcal{P}, \mathcal{I}, \mathcal{H}, \mathcal{R}).$$

Thus the reduced architecture exists.

Theorem B2. Feynman graph to GFAT architecture

Every finite Feynman graph determines a reduced Process Green-Function Architecture.

Proof.

A finite Feynman graph contains edges and vertices. Edges are propagators and therefore define propagation. Vertices define interactions. The first cycle space defines holonomy. Nested insertions and Dyson-type expansions define recursion. Hence every finite Feynman graph determines a reduced GFAT process architecture.

Theorem B3. Reduced multiplicity invariance

If two process architectures possess the same tuple

$$(a_p, b_p, c_p, d_p),$$

then they possess the same process Architectural Multiplicity.

Proof.

The multiplicity depends only on the tuple. Therefore, equal tuples imply equal multiplicities.

Appendix C. Experimental Validation Tables

C1. Decay-energy tests

Decay	GFAT quantity tested	Status
$\mu \rightarrow e\nu\nu$	Q_{GFAT}	close
$\pi^0 \rightarrow \gamma\gamma$	Q_{GFAT}	close
$K^+ \rightarrow \mu\nu$	Q_{GFAT}	promising
$\tau \rightarrow e\nu\nu$	Q_{GFAT}	needs correction
$n \rightarrow p e \nu$	charge/isospin correction required	

C2. Lifetime tests

Class	Preferred GFAT formula	Status
Pure leptonic weak decay	$\Pi_W^2 \rho(Q)$	promising
W leptonic decay	$\Pi_W^2 \rho(Q)$	consistent with lepton universality
Hadronic weak decay	$\Pi_W^2 \rho(Q) S_{\text{selection}}$	needs hadronic factors
Strong decay	$\Pi_{\text{strong}}^2 \rho(Q) S_{\text{selection}}$	preliminary
Radiative meson decay	$\Pi_{\text{EM}}^2 M^3 S_{\text{mixing}}$	promising for pseudoscalars

Appendix D. The Feynman–GFAT Correspondence

A formal mapping may be written as follows.

Let D be a finite Feynman diagram.

Define

$$\mathcal{A}(D) = (E(D), V(D), H_1(D), \mathcal{R}(D)).$$

Here

$$E(D)$$

is the set of propagator edges,

$$V(D)$$

is the set of interaction vertices,

$$H_1(D)$$

is the first homology or cycle space of the graph,

and

$$\mathcal{R}(D)$$

is the recursive insertion structure.

Then

$$\begin{aligned} E(D) &\rightarrow \mathcal{P}, \\ V(D) &\rightarrow \mathcal{I}, \\ H_1(D) &\rightarrow \mathcal{H}, \\ \mathcal{R}(D) &\rightarrow \mathcal{R}. \end{aligned}$$

Thus

$$D \rightarrow \mathcal{A}_{\text{process}}.$$

The Feynman diagram gives the perturbative amplitude.

The GFAT process architecture gives the structural decomposition of the same physical process.

Appendix E. Width Comparison Table

The experimental widths are taken from PDG-type particle data; GFAT particle energies and tuples are taken from the attached GFAT energy manuscript. The Feynman/Standard-Model column denotes the conventional amplitude-based prediction or phenomenological standard value using known couplings and phase-space factors.

Table E1. Leptonic and weak decays

Process	Γ_{GFAT}	$\Gamma_{\text{Feynman/SM}}$	Γ_{exp}	Comment
$\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$	$4.55 \times 10^5 \text{ s}^{-1}$	$4.55 \times 10^5 \text{ s}^{-1}$	$4.55 \times 10^5 \text{ s}^{-1}$	Calibration channel
$\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau$	$6.32 \times 10^{11} \text{ s}^{-1}$	$\sim 6.1 \times 10^{11} \text{ s}^{-1}$	$\sim 6.1 \times 10^{11} \text{ s}^{-1}$	Good agreement
$\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau$	$6.15 \times 10^{11} \text{ s}^{-1}$	$\sim 6.0 \times 10^{11} \text{ s}^{-1}$	$\sim 6.0 \times 10^{11} \text{ s}^{-1}$	Good agreement
$W \rightarrow e\nu$	equal leptonic width expected	$\sim 0.226 \text{ GeV}$	$\sim 0.226 \text{ GeV}$	Consistent with lepton universality
$W \rightarrow \mu\nu$	equal leptonic width expected	$\sim 0.226 \text{ GeV}$	$\sim 0.226 \text{ GeV}$	Consistent with lepton universality
$W \rightarrow \tau\nu$	near-equal, phase-space corrected	$\sim 0.226 \text{ GeV}$	$\sim 0.225 \text{ GeV}$	Consistent with lepton universality

Appendix E2. Hadronic-source weak decays

Calibrated to $\pi^+ \rightarrow \mu^+ \nu_\mu$.

Decay	Γ_{exp}	GFAT architecture-only	GFAT + quotient	Better model
$\pi^+ \rightarrow \mu^+ \nu$	$3.84 \times 10^7 \text{ s}^{-1}$	3.84×10^7	3.84×10^7	calibration
$K^+ \rightarrow \mu^+ \nu$	5.13×10^7	6.79×10^8	5.42×10^7	quotient
$D^+ \rightarrow \mu^+ \nu$	3.60×10^8	2.81×10^9	1.56×10^7	neither
$D_s^+ \rightarrow \mu^+ \nu$	1.09×10^{10}	2.95×10^9	1.49×10^7	neither
$D_s^+ \rightarrow \tau^+ \nu$	1.09×10^{11}	2.88×10^{10}	4.09×10^{10}	quotient partly
$B^+ \rightarrow \tau^+ \nu$	6.65×10^7	1.77×10^{12}	3.50×10^{11}	neither

Conclusion for E2: the quotient works impressively for K/π , but it is not universal.

Appendix E3. Strong hadronic decays

Calibrated to $\rho \rightarrow \pi\pi$, using p^3 -type two-body phase space.

Decay	Γ_{exp}	GFAT phase/architecture	GFAT + quotient	Better model
$\rho \rightarrow \pi\pi$	149.1 MeV	149.1	149.1	calibration
$K^* \rightarrow K\pi$	47.3 MeV	75.3	712.7	no quotient
$\phi \rightarrow KK$	4.25 MeV	6.44	583.3	no quotient
$\Delta \rightarrow N\pi$	117 MeV	36.9	661.4	no quotient
$\Sigma^* \rightarrow \Lambda\pi$	36 MeV	27.7	555.5	no quotient
$\Xi^* \rightarrow \Xi\pi$	9.1 MeV	11.5	262.5	no quotient

Summary for Appendix E

The current best-supported GFAT width law is

$$\Gamma_{\text{GFAT}} = \Gamma_0 g_{\text{arch}}^2 \Pi_{\text{arch}}^2 \rho(Q_{\text{GFAT}}) S_{\text{selection}}.$$